

```
Full Rank Matrix R:
[[-5.09   -6.27   -0.58]
 [ 0.      -1.27   -4.96]
 [ 0.        0.      0.15]]
Rank 3
```

Observation: The rank is 3, and all diagonal values are significantly different from zero.

Case B: Rank deficient matrix

```
# Rank deficient 3x3 matrix
X_def = np.array([
    [1, 2, 3],
    [2, 4, 6],
    [3, 6, 9]
])

Q_def, R_def = np.linalg.qr(X_def)

print("\nRank Deficient Matrix R:")
print(R_def)
print("Rank:", np.linalg.matrix_rank(X_def))

Rank Deficient Matrix R:
[[-3.74   -7.48  -11.22]
 [ 0    1.98*10-15  3.97*10-15]
 [ 0         0    -3.94*10-31]]
Rank: 1
```

Observation: Here you can see that two of the diagonal elements are almost 0 (1.98×10^{-15} and -3.94×10^{-31}). So the rank is 1.

```
r11 is -3.74 ≠ 0
r22 ≈ 0
r33 ≈ 0
```

In floating-point arithmetic, exact zero rarely appears. Values close to machine precision ($\approx 10^{-15}$) are treated as numerical zeros.

Since only the first diagonal entry of R is significantly non-zero, the matrix has rank 1. This implies that only the first column of X is linearly independent, and the remaining two columns are linear combinations of it.

Case C: Real-world application: NASA Turbofan dataset

To demonstrate this at scale, we apply QR decomposition to the NASA C-MAPSS Turbofan Engine Degradation Dataset.

Now let's take a real database from Kaggle.com at <https://www.kaggle.com/datasets/bishals098/nasa-turbofan-engine-degradation-simulation?resource=download&select=>

train_FD001.txt:

```
import numpy as np
import pandas as pd

# Load NASA dataset
df = pd.read_csv("train_FD001.txt", sep=" ", header=None)

# Keep only sensor columns (columns 0 to 25)
X = df.iloc[:, :26].values # 26 sensor columns

print("Shape:", X.shape)

# QR decomposition
Q, R = np.linalg.qr(X)

rank = np.linalg.matrix_rank(X)

print("Rank:", rank)
print("Number of columns:", X.shape[1])
print("Redundant columns:", X.shape[1] - rank)

print("\nAbsolute diagonal of R:")
print(np.abs(np.diag(R)))

Shape: (20631, 26)
Rank: 20
Number of columns: 26
Redundant columns: 6
```

Absolute diagonal of R:

```
[8.50622114e+03 1.21149103e+04 3.14141179e-01 4.20835556e-02
 5.77042252e+03 3.97731246e-10 5.99454189e+01 6.65820338e+02
 7.72601912e+02 3.36614676e-13 1.97019588e-01 7.08089878e+01
 5.97296214e+00 2.45961341e+03 4.41360662e-14 1.67406676e+01
 4.74133747e+01 4.84852689e+00 6.59181616e+02 2.98356370e+00
 1.91411652e-16 1.39192962e+02 3.10864101e-11 6.71705315e-13
 1.48371815e+01 8.80322651e+00]
```

Results and interpretation: The output reveals that out of 26 columns, 6 are redundant. When looking at the diagonal of R , several values appear in the range of 10^{-10} to 10^{-16} .

These near-zero values indicate features that provide no new information to a model. By identifying these via QR decomposition, data scientists can perform feature selection with mathematical certainty, ensuring more stable and efficient machine learning pipelines. 

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